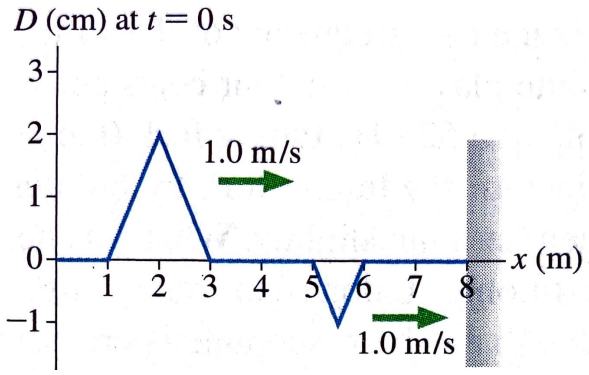
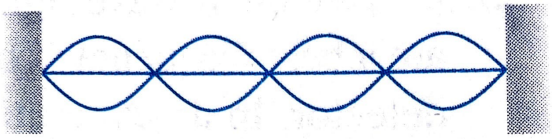
1. The figure below is a snapshot graph at t = 0 s of two waves moving to the right at 1.0 m/s. The string is fixed at x = 8 m. Draw four snapshot graphs, stacked vertically, showing the string at t = 2,4,6, and 8 s.



2. The figure shows a standing wave on a 2.0-m-long string that has been fixed at both ends and tightened until the wave speed is 40 m/s.



A. What is its frequency?

B. How many nodes/ antinodes will there be if the frequency is doubled?

C. If the Tension is increased by a factor of four, at what frequency will the string continue to oscillate so the standing wave looks like the one on the right?

3. Lasers set up standing waves of light in their cavities. A Carbon Dioxide laser has a cavity length of 53 cm and oscillates in the n= 100,000 mode. What are the wavelength and frequency of the laser?

4. Write down the equation for an n = 3 standing wave with a frequency of 250 hz, on a 1.5 m long piece of string. The full amplitude is 0.5 m.

5. Produce the following animation in mathematica over a full period. An n = 2 standing wave is created on a string under 1000 N of tension, with a linear mass density of $\mu = 0.1$ kg/m. Produce animations for n=3 and 4 as well.